

SMRITI SINGH

+91 9471723727 | smriti.shaurya22@gmail.com | [linkedin.com/in/smriti-singh-83541b267](https://www.linkedin.com/in/smriti-singh-83541b267) | github.com/SmrSingh

EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science Engineering (AI & ML)

June 2023 – Present

CGPA: 9.24/10

EXPERIENCE

Tata Steel Ltd

Intern

June 2026 – August 2026

Jamshedpur, India

- Built an end-to-end machine learning pipeline to predict molten steel temperature drop between the Ladle Furnace and the casting measurement point using historical production data.
- Cleaned and validated 3,287+ industrial records by handling missing values, duplicates and outliers using domain-based checks.
- Created 6 domain-specific features from process temperatures and timings to improve the model inputs.
- Tuned a Random Forest Regressor using RandomizedSearchCV and achieved an MAE of 6.9°C, i.e. predictions were off by about 7°C on average.
- Developed modular code for preprocessing, training and batch prediction, and a Streamlit app that gives engineers instant predictions of temperature drop and recommended LF exit temperature.

PROJECTS

MediMind AI – Intelligent Clinical Decision Support System

FastAPI, React, ChromaDB, Gemini, RAG

July 2026 – Present

Live Demo

- Built a web application where users upload CBC and LFT lab report PDFs and get an AI-generated summary of their results, using FastAPI, React, ChromaDB and Google Gemini.
- Developed a PDF processing pipeline to extract lab parameters with their values, units and reference ranges from the reports.
- Marked each value as High, Low or Normal by comparing it with its reference range.
- Implemented a RAG pipeline with ChromaDB and Sentence Transformers over medical guideline documents to explain abnormal values with source references.
- Added a chat feature to ask questions about the uploaded report, with REST APIs and a React interface for the full flow.

Multi-Agent AI Research Assistant

LangChain, Gemini, Streamlit

May 2026 – June 2026

Live Demo

- Built a multi-agent system with Search, Writer and Critic agents that turns a single topic into a researched report, automating search, reading, writing and review.
- Created custom tools using Tavily Search API and BeautifulSoup to find and read the top 3 web sources for each topic.
- Wrote prompts so the Writer agent uses only the collected sources and lists the source links, making reports easy to verify.
- Added a Critic agent that reviews the report, gives a score out of 10 and suggests improvements.
- Developed a Streamlit interface with one-click report download and deployed the application on Render.

TECHNICAL SKILLS

Languages: C++, SQL

Technologies: FastAPI, REST APIs, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Prompt Engineering

Core CS: DSA, OOP, DBMS, Operating Systems

CO-CURRICULAR ACTIVITIES

- Co-Lead, Design Team, IoT Club, VIT Bhopal – Led design initiatives and collaborated on organizing technical events.
- Won two Silver Medals in the Bench Press and Squat events at the VIT Sports Fest.

CERTIFICATIONS

- Applied Machine Learning in Python – University of Michigan (Coursera)
- Machine Learning in collaboration with Google for Developers